

GPU-Accelerated Gravitational Wave Parameter Estimation and Hubble Constant Inference

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Abstract

The primary scientific aim of this dissertation is the inference of the Hubble constant (H_0) using gravitational wave (GW) astronomy. The advent of GW astronomy has opened new avenues for probing fundamental physics and cosmology; however, parameter estimation for GW sources presents significant computational challenges, particularly with the anticipated data volumes from incoming third-generation (3G) GW detectors. These challenges hinder rapid and precise H_0 measurements from standard sirens. To address this, I explored the acceleration of GW parameter estimation using Graphics Processing Units (GPUs) coupled with nested sampling algorithms. I developed and validated a GPU-accelerated pipeline based on the JAX stack, demonstrating substantial improvements in runtime efficiency compared to standard CPU implementations. My analysis reproduced the H_0 posterior distribution derived from the first observed bright siren, GW170817, as presented by Abbott et al. (2017b), using this GPU pipeline. Furthermore, I critically examined the methodology employed in that work for assessing prior sensitivity, specifically the reweighting technique applied to achieve a flat-in-redshift prior. By directly sampling alternate priors using my JAX-based pipeline, I demonstrate that the reweighting approach can be misleading due to limitations in sample coverage. My results highlight the potential of GPU acceleration to enable rapid and robust GW parameter estimation, crucial for multi-messenger follow-up and for leveraging the discovery potential of 3G detectors, including the potential to resolve the Hubble tension through population analysis of bright sirens.

1 Introduction

The inference of the Hubble constant (H_0) stands as a pivotal challenge in modern cosmology, and gravitational wave (GW) astronomy offers a promising independent pathway to address it. Inaugurated by the landmark detection of GW150914 (Abbott et al., 2016c), GW astronomy has rapidly evolved into a cornerstone of modern astrophysics. One of its most compelling applications lies in providing measurements of H_0 that are independent of traditional electromagnetic (EM) methods. The current discrepancy between early-universe measurements of H_0 (e.g., from the Cosmic Microwave Background (CMB) by Planck (Ade et al., 2016)) and late-universe measurements (e.g., using Type Ia supernovae calibrated via the cosmic distance ladder by SH0ES (Riess et al., 2016)) constitutes the Hubble Tension¹, a significant challenge to the standard Λ CDM cosmological model. My work described in this paper is primarily focused on developing and applying advanced computational techniques to contribute to H_0 measurements using GWs.

Gravitational wave sources, particularly compact binary coalescences (CBCs) involving neutron stars or black holes, can serve as ‘standard sirens’, a term coined by (Schutz, 1986). The GW signal directly encodes the luminosity distance (d_L) to the source. If an EM counterpart is detected (a ‘bright siren’), the host galaxy can be directly identified spectroscopically, providing an independent redshift (z) measurement. Combining d_L and z allows for a direct estimate of H_0 via the Hubble-Lemaître law, bypassing the traditional cosmic distance ladder and its associated systematic uncertainties. The

¹The Hubble tension refers to the $4 - 6\sigma$ disagreement between CMB (Ade et al., 2016) and SNaE (Riess et al., 2016) measurements of H_0 .

binary neutron star (BNS) merger GW170817, with its associated gamma-ray burst and kilonova (Abbott et al., 2017c), provided the first, and to date only, such multi-messenger measurement of H_0 . With the advent of third-generation (3G) detectors, which are expected to detect approximately 10^5 CBC signals each year (Amaro-Seoane et al., 2017), we will eventually be able to perform population analyses of bright sirens to obtain a highly constrained measurement of H_0 , potentially resolving the Hubble tension.

However, the process of extracting physical parameters from GW data – known as parameter estimation – is computationally intensive. This presents a major bottleneck to fully exploiting GWs for cosmological inference. Standard GW parameter estimation pipelines rely on Bayesian inference, typically employing stochastic sampling algorithms like Nested Sampling (NS) or Markov Chain Monte Carlo (MCMC) to explore the high-dimensional parameter space (15-17 parameters for the latest waveform models as discussed in Colleoni et al. (2025)) and map the posterior probability distributions. These parameters include approximately 8 intrinsic parameters (e.g., masses, spins), 7 extrinsic parameters (e.g., sky location, phase), and additional parameters describing further effects such as tidal deformability in neutron stars.

Current implementations of parameter estimation, such as those within the widely used Bilby framework (Ashton et al., 2019), predominantly run on Central Processing Units (CPUs). The standard parameter estimation run used by the LIGO-Virgo-KAGRA (LVK) Collaboration is extremely computationally expensive and employs few acceleration methods. For example, Colleoni et al. (2025) analysed GW170817 using default settings, as per the LVK analysis in Abbott et al. (2019a), and reported a runtime of approximately 9700 CPU hours. This computational cost will not scale with the impending era of 3G ground-based detectors, such as the Einstein Telescope² and Cosmic Explorer³, along with the space-based Laser Interferometer Space Antenna (LISA)⁴.

These 3G detectors promise a large influx of data, as they possess significantly greater sensitivity—an order of magnitude improvement in signal-to-noise ratio (SNR) compared to current 2G detectors—and will operate over wider frequency bands. Furthermore, these detectors will observe significantly longer duration signals; for instance, LISA, operating in the milliHertz band, will observe Supermassive Black Hole Binary mergers with inspiral phases ranging from weeks to years, and merger and ring-down phases between hours and days. These 3G detectors will detect overlapping GW signals from numerous sources simultaneously. Performing parameter estimation on this volume of data with the standard LVK CPU-based parameter estimation pipelines will be computationally prohibitive, potentially requiring on the order of $10^9 - 10^{12}$ CPU core hours to analyse a one-month-observation catalogue (Hu & Veitch, 2025).

Rapid parameter estimation is crucial for multi-messenger astronomy, a key pathway to precise H_0 measurements from bright sirens. Identifying EM counterparts like kilonovae or gamma-ray bursts requires prompt localisation of the GW source on the sky, typically within hours. The time sensitivity of these follow-up observations necessitates parameter estimation pipelines significantly faster than current capabilities allow, particularly in the 3G era. Failure to achieve this will hinder our ability to exploit the full scientific potential of multi-messenger observations, including the discovery and analysis of bright sirens needed for precision cosmology.

This project directly addresses this critical need for faster GW parameter estimation as a means to improve H_0 measurements. I proposed and implement a parameter estimation pipeline leveraging the massively parallel architecture of GPUs, utilising the JAX framework (Bradbury et al., 2018). My primary goal was to demonstrate that GPU acceleration can drastically reduce parameter estimation runtimes while maintaining accuracy, thereby enabling rapid analysis suitable for the 3G era and facilitating timely multi-messenger follow-up. I specifically focused on applying this accelerated pipeline to the cosmological inference of H_0 using the bright siren GW170817, closely following the CPU-based analysis of Abbott et al. (2017b).

²Further information available at <https://www.et-gw.eu>

³Further information available at <https://cosmicexplorer.org>

⁴Further information available at <https://www.lisamission.org>

The novelty of my work lies in the development and application of a JAX-based GPU implementation of Bayesian parameter estimation, specifically tailored for joint GW source parameter and H_0 inference using nested sampling. I build upon previous work by [Wong et al. \(2023\)](#) in the form of the Jim Python library, and implemented the nested slice sampler developed by [Yallup et al. \(2025\)](#). My aim was not only to accelerate parameter estimation but also to create a more lightweight, modular, and understandable system compared to established, complex frameworks like Bilby. The increasing availability and cost-effectiveness of GPU resources, including cloud computing platforms, further motivates this shift towards GPU-centric analysis, alongside the inherent framework advantages I discuss in later sections.

This report is structured as follows. Section 2 provides the theoretical background on gravitational waves, standard sirens and their role in H_0 inference, Bayesian inference, parameter estimation techniques including nested sampling, and the computational challenges involved. Section 3 details my methodology for incorporating H_0 inference, the specific implementation choices I made, the waveform models I employed and the overall development of the GPU-accelerated pipeline. Section 4 presents the validation of my pipeline using the benchmark event GW150914, analysis of GW170817, the derivation of H_0 , a critical re-evaluation of prior sensitivity analysis from [Abbott et al. \(2017b\)](#), and a performance comparison between my GPU pipeline and other CPU-based methods. Finally, Section 5 summarises my key findings and their implications for future H_0 measurements.

2 Theory and Background

This section provides the necessary theoretical context for understanding the methods I employed and the results I obtained in this paper. I will briefly cover the fundamentals of gravitational waves, landmark detections relevant to my work, the concept of standard sirens and their crucial role in cosmological measurements (particularly of H_0), Bayesian inference as applied to GW data analysis, the nested sampling algorithm, and the computational challenges that motivate my development of GPU-accelerated techniques.

2.1 Gravitational Waves

Gravitational waves are ripples in the fabric of spacetime, predicted by Einstein’s theory of General Relativity (GR). They are generated by accelerating massive objects, with the most detectable sources for current instruments being CBCs – systems of two orbiting black holes, neutron stars, or one of each, spiralling inwards and eventually merging. As the binary orbits, it emits GWs that carry away energy and angular momentum, causing the orbit to shrink and the components to accelerate. This process generates a characteristic ‘chirp’ signal, where the frequency and amplitude of the GWs increase over time, culminating in the final merger and subsequent ringdown phase as the newly formed object settles. The precise shape, or waveform, of this signal encodes detailed information about the source, including the masses and spins of the component objects, their orbital orientation, and, crucially for H_0 measurements, the luminosity distance to the source.

2.2 Landmark Gravitational Wave Events

Two specific GW events serve as primary case studies throughout my project: GW150914 for pipeline validation and GW170817 for H_0 inference.

2.2.1 GW150914: The First Direct Detection

Detected on September 14, 2015, GW150914 marked the first direct observation of gravitational waves and the first observation of a binary black hole (BBH) merger ([Abbott et al., 2016c](#)). This event originated from the merger of two black holes, with approximate masses of $\sim 36M_\odot$ and $\sim 29M_\odot$, at a luminosity distance of approximately 410 Mpc.

Its detection provided compelling confirmation of GR in the strong-field regime and validated the technical capabilities of GW detectors. Due to the extensive analyses performed by the LVK, GW150914 serves as a crucial benchmark for testing and validating GW parameter estimation pipelines. In Section 4.1, I use this event to verify the capability of my GPU-based pipeline to recover known results.

2.2.2 GW170817: Multi-Messenger Astronomy

Observed on August 17, 2017, GW170817 was the first detection of gravitational waves from a binary neutron star (BNS) merger (Abbott et al., 2017a). Crucially, it was accompanied by EM counterparts across the spectrum, including a short gamma-ray burst (GRB 170817A) detected 1.7 seconds after the GW merger time by the Fermi Gamma-ray Space Telescope, followed by an optical/infrared kilonova (AT 2017gfo) discovered 11 hours later in the galaxy NGC 4993. This marked the dawn of multi-messenger astronomy with GWs.

GW170817 remains the only confirmed multi-messenger GW event to date and serves multiple vital roles for this work. Firstly, it underscores the importance of rapid parameter estimation: the initial GW sky localisation, refined using data from the LIGO-Hanford (H1), LIGO-Livingston (L1), and Virgo (V1) detectors, was essential for guiding the telescope searches that found the EM counterparts. This need for speed will only intensify with 3G detectors. Secondly, as the only observed bright siren (a GW event with an identified host galaxy providing redshift information), it enabled the first standard siren measurement of the Hubble constant, yielding $H_{0,\text{LVK}} = 70.0^{+12.0}_{-8.0} \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Abbott et al., 2017b). This event forms the basis of my H_0 analysis in Sections 3.2 and 4.3.

2.2.3 Gravitational Wave Data

The GW strain data (time-series data representing the fractional change in detector arm length) I used in this project are publicly available from the Gravitational Wave Open Science Center (GWOSC) (Abbott et al., 2023). GWOSC hosts datasets around confirmed LVK detections, typically including detector strain data and associated metadata such as noise power spectral densities (PSDs).

For comparison and validation of my pipeline, I utilise ‘standard’ parameter estimation results produced by the LVK collaboration, which are sourced from Gravitational-Wave Transient Catalog (GWTC) releases (Abbott et al., 2021). These catalogues contain curated lists of GW events with associated parameter estimates, often presented as median values and 90% confidence intervals, along with full posterior samples derived using LVK analysis pipelines. Parameter names typically follow the standard laid out in PESummary⁵.

Figure 1 shows time-frequency representations (Q-transforms) of the GW170817 strain data from the H1, L1, and V1 detectors around the time of the merger. The characteristic chirp signal is clearly visible in the H1 and L1 data but is sub-threshold in V1. Despite this, the data from V1 still contributed significantly to constraining the GW source’s sky location and polarisation (Abbott et al., 2017a). The L1 data shown includes both the original data containing a short instrumental glitch near the merger time, and the reprocessed data with the glitch removed (Abbott et al., 2023). My project uses the cleaned L1 data for analysis.

⁵More information at: <https://lscsoft.docs.ligo.org/pesummary/stable/gw/parameters.html>

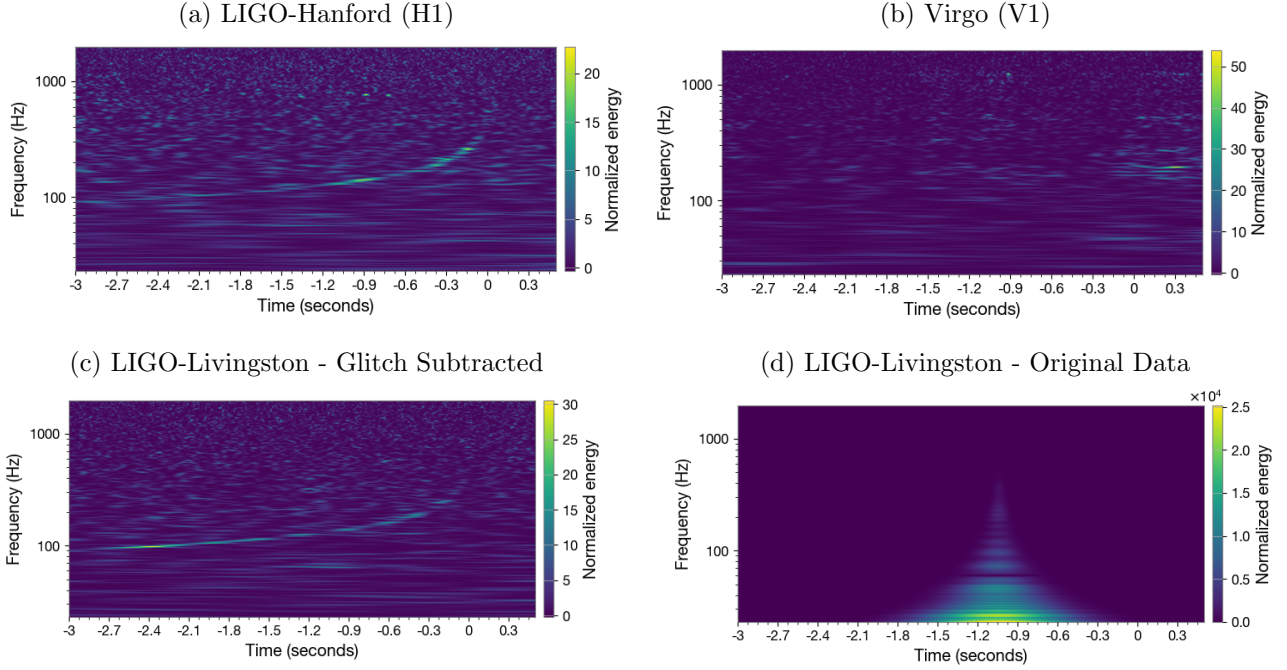


Figure 1: Time-frequency representations (Q-transforms) of the LIGO-Hanford (H1), LIGO-Livingston (L1), and Virgo (V1) detector strain data around GW170817 (Trigger time: 17 Aug 2017 12:41:04.43 UTC). Data sourced from GWOSC (Abbott et al., 2023). The colour scale represents normalised energy. The characteristic chirp signal is below the noise floor in V1 (panel b). Panel (c) shows the data after removal of an instrumental glitch; panel (d) shows the original data containing the glitch artifact.

2.3 Standard Sirens and the Hubble Constant

A key promise of GW astronomy, and the central focus of my project’s application, is its ability to provide independent cosmological measurements. Unlike electromagnetic observations, which often rely on a complex and systematically uncertain cosmic distance ladder to measure distances, GWs offer an independent route.

2.3.1 Standard Candles and the Distance Ladder

Traditional cosmology measures distances using the cosmic distance ladder (Riess et al., 2022). This involves calibrating standard candles – astronomical objects of known intrinsic brightness (luminosity) – whose distances can be inferred from their observed apparent brightness. Nearby distances are measured directly using parallax. These measurements then calibrate primary candles like Cepheid variable stars, which in turn calibrate secondary candles like Type Ia supernovae that are visible at cosmological distances. While powerful, this method relies on accurate calibration at each step, and potential systematic errors can propagate up the ladder and accumulate.

2.3.2 Standard Sirens

Gravitational wave sources, particularly CBCs, can act as standard sirens. GR predicts the intrinsic amplitude of the GW signal emitted by a coalescing binary. By measuring the observed GW strain amplitude, one can directly infer the luminosity distance (d_L) to the source. This measurement is independent of EM distance indicators and their calibrations. However, there is a significant degeneracy between the inferred luminosity distance and the inclination angle (ι) of the binary’s orbital plane relative to the line of sight. Both parameters affect the observed waveform amplitude (Usman et al., 2019), which leads to significant uncertainty in d_L if ι is poorly constrained and vice versa. Additional information, such as constraints from EM counterparts (e.g., GRB jets) can help break this degeneracy.

GW events are classified as ‘bright’ or ‘dark’ sirens:

- Bright sirens are GW events for which an EM counterpart is detected, allowing for direct identification of the host galaxy. Measuring the host galaxy’s redshift via spectroscopy then allows for a direct estimate of the Hubble constant using the distance-redshift relation for that single event. GW170817 is the prime example.
- Dark sirens are GW events without a detected EM counterpart. Current dark siren methods for H_0 inference can be broadly split into galaxy catalogue methods and population methods (also known as ‘spectral sirens’, discussed by [Gray et al. \(2023\)](#)). The catalogue method statistically infers H_0 by correlating the GW localisation volume with galaxy catalogues. While a single dark siren provides weak constraints, combining information from many events can yield a precise H_0 measurement, as incorrect host associations are statistically averaged out.

2.3.3 The Hubble Constant and Tension

The Hubble constant, H_0 , quantifies the current expansion rate of the Universe. Measuring it accurately is fundamental to cosmology; however, a significant tension exists between measurements from the early Universe and the late Universe (see [Di Valentino et al. \(2021\)](#) for a review). Early universe analyses of the CMB by the Planck satellite yield $H_{0,\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ([Ade et al., 2016](#)), assuming the standard Λ CDM model. In contrast, late universe measurements using the cosmic distance ladder (e.g., the SH0ES project) find $H_{0,\text{SH0ES}} = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ([Riess et al., 2016](#)). This discrepancy challenges the standard cosmological model, and motivates the search for independent measurement techniques.

Standard sirens offer a completely independent method to measure H_0 , potentially arbitrating this tension. For a nearby bright siren like GW170817 (with redshift $z \approx 0.01$), within a standard flat Λ CDM cosmology, the relationship between distance and redshift is well approximated by the Hubble-Lemaître law:

$$v_H = H_0 d_L \quad (1)$$

where v_H is the recession velocity of the GW source (approximated by its host galaxy) due to cosmic expansion (the Hubble flow). The observed redshift z of the host galaxy is related to the total recessional velocity (v_r), which includes its peculiar motion (v_p) relative to the Hubble flow:

$$v_r = cz \approx v_H + v_p = H_0 d_L + v_p \quad (2)$$

where c is the speed of light. Therefore, H_0 can be estimated from a bright siren as:

$$H_0 \approx \frac{cz - v_p}{d_L} \quad (3)$$

The luminosity distance d_L is inferred from the GW signal amplitude (degenerately with inclination ι , as discussed). The redshift z is measured from the host galaxy’s spectrum. Correcting the observed recessional velocity cz for the peculiar velocity v_p (which must be estimated or modelled) gives a value for the Hubble velocity v_H , and thus allows for an estimate of H_0 .

Achieving higher precision with future bright sirens requires accurately modelling d_L , ι , and v_p within a Bayesian framework, alongside rapid parameter estimation to enable the identification of EM counterparts. As the number of bright siren detections increases with future detectors, this method promises an increasingly precise and independent measurement of H_0 .

2.4 Bayesian Inference in Gravitational Wave Astronomy

Bayesian inference provides a rigorous framework for parameter estimation in the presence of noise and uncertainty, making it the standard approach in GW data analysis.

Given observed strain data $\mathcal{D}$ from one or more GW detectors, and a theoretical waveform model $\mathcal{M}$ (detailed in Section 3.1.1), that depends on a set of source parameters Θ (e.g., masses, spins,

distance), we aim to infer the posterior probability distribution $\mathcal{P}(\Theta|\mathcal{D}, \mathcal{M})$. This posterior represents our knowledge of the parameters after observing the data. Bayes' theorem states:

$$\mathcal{P}(\Theta|\mathcal{D}, \mathcal{M}) = \frac{\mathcal{L}(\mathcal{D}|\Theta, \mathcal{M})\pi(\Theta|\mathcal{M})}{\mathcal{Z}(\mathcal{D}|\mathcal{M})} \quad (4)$$

where:

- $\mathcal{P}(\Theta|\mathcal{D}, \mathcal{M})$ is the posterior probability distribution of the parameters Θ , given the data $\mathcal{D}$ and the model $\mathcal{M}$. This is the primary output of the inference process.
- $\mathcal{L}(\mathcal{D}|\Theta, \mathcal{M})$, often written simply as $\mathcal{L}(\Theta)$, is the likelihood function. This quantifies the probability of observing the data $\mathcal{D}$ given a specific set of parameter values Θ and model $\mathcal{M}$. In GW analysis, this typically involves comparing the predicted waveform generated for parameters Θ with the observed data, x_{GW} .
- $\pi(\Theta|\mathcal{M})$, often written as $\pi(\Theta)$, is the prior probability distribution. This encodes our prior knowledge or assumptions about the parameters Θ before observing the data. Priors are chosen based on astrophysical expectations, theoretical constraints, or to be uninformative over a given range.
- $\mathcal{Z}(\mathcal{D}|\mathcal{M}) = \int \mathcal{L}(\mathcal{D}|\Theta, \mathcal{M})\pi(\Theta|\mathcal{M})d\Theta$ is the Bayesian evidence (or marginal likelihood), which acts as a normalisation constant in Equation 4. It represents the total probability of observing the data $\mathcal{D}$ given the model $\mathcal{M}$, marginalised over all possible parameter values. The evidence is crucial for model selection, allowing comparison of the relative probabilities of different waveform models or hypotheses.

2.5 Existing Frameworks: Bilby

Bilby (Ashton et al., 2019) is a widely adopted, open-source Python library for Bayesian inference, which has become the standard for parameter estimation analysis. It provides a user-friendly interface to various sampling algorithms, standard GW likelihood functions, prior definitions, and waveform generation libraries (primarily LALSuite (LIGO Scientific Collaboration et al., 2018)). Bilby is primarily designed for execution on CPUs.

In this study, Bilby serves as the primary benchmark for parameter estimation, with the standard LVK posteriors generally being obtained from Bilby. While powerful and flexible, its layered structure and reliance on external CPU-bound libraries limits performance and makes direct GPU porting complex. This inherent CPU-centric design limits its performance scalability for the large-scale analyses anticipated with 3G detectors.

2.6 Nested Sampling

Evaluating the posterior distribution $\mathcal{P}(\Theta|\mathcal{D}, \mathcal{M})$ and evidence $\mathcal{Z}$ typically requires numerical methods, as analytical solutions are rarely feasible for complex models and high-dimensional parameter spaces. Nested Sampling (NS), introduced by Skilling (2004); Skilling (2006), is a Monte Carlo algorithm designed specifically for Bayesian computation. It calculates both the posterior samples and the Bayesian evidence $\mathcal{Z}$. NS is widely used in astrophysics and cosmology and is considered the gold standard for GW parameter estimation by the LVK Collaboration.

The general algorithm proceeds as follows:

Step 1: Initialisation A set of N ‘live’ points is initialised by drawing samples from the prior distribution $\pi(\Theta)$. The likelihood $\mathcal{L}(\Theta)$ is evaluated for each point.

Step 2: Iteration At each iteration i , the live point with the *lowest* likelihood value, $\mathcal{L}_{\text{worst}} = \mathcal{L}_i$, is identified and removed from the live set. It is added to a list of ‘dead’ points. A new point is drawn from the prior, subject to the constraint that its likelihood must be *higher* than $\mathcal{L}_{\text{worst}}$, and replaces the discarded point.

Step 3: Evidence Accumulation At each iteration i , the volume of the prior space X_i shrinks with a compression factor $t_i = X_i/X_{i-1} \sim e^{-1/N}$. Thus the prior mass associated with the discarded point is $w_i = X_{k-1} - X_k$. This dead point contributes $\mathcal{L}_{\text{worst}} \times w_i$ to the evidence estimate $\mathcal{Z}$, hence the total evidence is estimated as $\mathcal{Z} \approx \sum_i \mathcal{L}_i w_i$.

Step 4: Termination The process continues until a stopping criterion is met, indicating the remaining live points contribute negligibly to the evidence integral. In my code, I use the difference between the log-evidence of the live points and the current log-evidence, with the criteria $\mathcal{Z}_{\text{live}} - \mathcal{Z} < -3$.

Step 5: Posterior Samples The sequence of discarded (dead) points, appropriately weighted by their contribution to the evidence ($p_i \propto \mathcal{L}_i w_i / \mathcal{Z}$), forms a set of samples from the posterior distribution $\mathcal{P}(\Theta|\mathcal{D}, \mathcal{M})$.

NS directly explores the likelihood landscape by moving towards regions of higher likelihood, making it effective even for posteriors with multi-modal structures. The ability to provide a direct estimate of the evidence $\mathcal{Z}$ is a significant advantage over other sampling methods.

However, NS can be computationally expensive. Step 2, finding a replacement point, typically requires running an auxiliary MCMC sampler, confined within the likelihood contour defined by $\mathcal{L} > \mathcal{L}_i$. This often necessitates many likelihood evaluations per iteration. Furthermore, while independent NS runs can be parallelised, the core algorithm has serial dependencies (finding the worst point, and sampling a new one). This makes efficient scaling on CPU architectures challenging for very high dimensions or computationally expensive likelihoods, which is a motivation for my work on GPU acceleration.

In this paper, the CPU-based comparisons use `Bilby` with the `PyPolyChord` (Handley et al., 2015a,b) nested sampler, while my GPU pipeline uses a JAX-based implementation of nested slice sampling developed by (Yallup et al., 2025).

Posterior distributions obtained from NS (or MCMC) are often visualised using ‘corner plots’ or ‘triangle plots’ which I use throughout Section 4. These plots display the 1D marginalised posterior distributions for each parameter, along the diagonals, and the 2D marginalised distributions (as contour plots) for each pair of parameters.

2.7 Acceleration Techniques

To mitigate the computational cost of likelihood evaluations, various acceleration techniques have been developed. In my project, I primarily focused on investigating the capabilities of the JAX ecosystem for hardware-level acceleration. Among existing algorithmic acceleration techniques, I also chose to implement and test heterodyning, or relative binning, as described by Zackay et al. (2018). This provides a complementary approach to other methods such as Reduced Order Quadrature (ROQ). Integrating further acceleration techniques with the pipeline I developed is a promising avenue for future work.

2.7.1 Heterodyning (Relative Binning)

The core idea behind heterodyning, or relative binning, as introduced by Cornish (2013); Cornish (2021); Zackay et al. (2018), is that during parameter estimation, a significant number of likelihood evaluations occur for parameter values close to the true or maximum likelihood values. In this region, the GW waveform changes relatively smoothly as a function of the parameters. This allows for approximations that reduce the computational cost.

By defining a fiducial high-likelihood waveform h_0 , often found from an initial quick optimisation, we may approximate full likelihood calculations. This allows the likelihood calculations to be performed on a coarser frequency grid, i.e. ‘binned’ frequencies.

2.7.2 GPU-Based Acceleration

GPUs possess fundamentally different hardware architectures compared to CPUs. A typical modern GPU contains thousands of relatively simple processing cores, optimised for performing the same operation in parallel on large streams of data. This contrasts with CPUs, which have fewer, more complex cores designed for general-purpose serial tasks and handling complex branching logic.

This massive parallelism makes GPUs exceptionally well-suited for tasks common in scientific computing, including GW parameter estimation. Key advantages relevant to my work include:

- The generation of waveforms involves repetitive calculations (e.g., evaluating phase and amplitude formulae) across a range of frequencies. This maps naturally to GPU parallelism, where each core can handle a frequency bin or a batch of waveforms simultaneously.
- Many numerical operations can be cast as matrix or vector (linear algebra) operations, for which GPUs provide highly optimised libraries (e.g., CUDA). The nested sampler I implemented (Yallup et al., 2025) leverages this. For example, the likelihood calculation involves inner products between data and templates, across multiple detectors. These operations can be significantly accelerated with matrix formulation, especially when evaluating likelihoods for batches of parameter points simultaneously, as is common within a NS algorithm.

Furthermore, GPUs typically offer significantly higher memory bandwidth than CPUs, which is advantageous when dealing with large datasets such as long-duration GW data from 3G detectors. Frameworks like JAX abstract away much of the low-level complexity of GPU programming, allowing researchers to write high-level code (often Python) that is automatically compiled and optimised for execution on GPUs. This simplifies the development process, making the power of GPUs more accessible. JAX’s features like automatic differentiation also open possibilities for gradient-based sampling methods like Hamiltonian Monte Carlo (HMC), although my work focuses on NS.

2.7.3 Other Techniques

Besides GPU acceleration and heterodyning, there are various other techniques being employed.

- Reduced Order Quadrature (ROQ) uses compressed representations of waveform families generated from Reduced Order Modelling (ROM) techniques. These provide accurate approximations of complex waveforms using a smaller number of parameters, allowing for fast likelihood evaluations (Canizares et al., 2015; Smith et al., 2016). ROQ then accelerates this further by reducing the calculations to linear and quadratic summations.
- Analytical marginalisation of certain parameters like coalescence phase and luminosity distance is commonly utilised by the LVK. These parameters are analytically integrated out from the likelihood, and are thus not explicitly sampled over. This reduces the dimensionality of the sampling. These parameters are then reconstructed during post-processing, once sampling has concluded.
- Machine learning methods such as Dingo (Dax et al., 2021) are seeing increased adoption and usage. These techniques often involve training neural networks to approximate the posterior distributions or likelihood functions. While these methods often require significant upfront training time, they are able to provide fast inference once trained. These techniques hold high potential for future applications with the incoming 3G detectors.

2.8 Computational Challenges

As highlighted in the Introduction, the computational demands of GW parameter estimation are substantial and are projected to grow significantly:

- Accurately modelling GW signals, especially for BNSs (which require incorporating tidal effects) or precessing BBHs (where spin-orbit interactions induce modulations), requires complex and

computationally expensive waveform models. These models will only increase in complexity as we include more subtle physical effects like eccentricity or higher-order waveform modes.

- CBC parameter spaces typically involve 13 – 17 dimensions. Exploring such spaces efficiently is challenging, as the volume grows exponentially with dimension. The posterior distribution often occupies a tiny fraction of the prior volume, requiring sophisticated stochastic sampling algorithms to draw representative samples from the underlying probability distribution.
- In the era of LISA and other 3G detectors, it is expected that signals from multiple GW sources will overlap in the time-frequency domain. Disentangling these sources requires more complex analysis techniques capable of joint parameter estimation for all sources simultaneously. This requires ‘global fit’ approaches that will dramatically increase the effective dimensionality and hence computational cost of the inference.
- Sensitivity upgrades to current detectors and future 3G detectors will increase the detection rate of GW events significantly. Coupled with the above challenges of complex waveforms, overlapping signals, and long signal durations, analysing 3G era events using standard parameter estimation techniques could require of order 10^{10} CPU hours (Hu & Veitch, 2025). While current acceleration methods significantly reduce this time cost to within 1000 CPU hours, the demand remains substantial.

These challenges collectively highlight the urgent need for more efficient computational methods. As discussed, the time-critical nature of multi-messenger astronomy, crucial for H_0 measurements, demands parameter estimation results within timescales of minutes to hours. This is a timescale currently difficult to achieve consistently, especially for complex analyses or lower SNR events, using traditional CPU-based approaches. Leveraging the architecture of GPUs provides a complementary and scalable path forward, especially given the increasing availability and computational power of GPU resources in High-Performance Computing (HPC) centres and commercial cloud platforms.

3 Methodology

In this section I present the different analysis models I used and detail the procedure for incorporating cosmological parameter estimation into the GW inference framework. I discuss the methods I employed in this project to design, implement, and validate the GPU-accelerated pipeline, describe my choice of computational tools, and outline the code development process.

3.1 Analysis Models, Priors and Waveforms

The choice of waveform model, priors, and overall analysis configuration impacts the results and computational expense of GW parameter estimation. In this section, I describe the specific waveforms used in my work, the different analysis models I set up for testing, and the priors employed.

3.1.1 Waveform Models

Gravitational waveform models are theoretical predictions of the observed GW signal $h(t)$, derived from GR. These often employ either phenomenological or effective-one-body (EOB) approximations. The accuracy of these models is crucial for unbiased and valid parameter estimation. My choice of waveforms was constrained by the selection of validated models within the `Ripple` (Edwards et al., 2023) library, which provides JAX-based waveform implementations suitable for GPU execution.

For the validation event GW150914 (a BBH merger), the LVK Collaboration’s parameter estimation results (Abbott et al., 2019b) utilised the `IMRPhenomXPHM` model. This accounts for precessing spins and higher order modes. However, `Ripple` does not currently offer full support for BBH precessing spin waveform models of this complexity that have undergone complete validation. While `Ripple` does include the precessing `IMRPhenomPv2` model, its sampling checks were still ongoing. Therefore I chose not to implement it and to instead used the fully validated `IMRPhenomD` (Khan et al., 2016)

model for GW150914. This BBH model assumes that the spins of the BHs are aligned with the orbital angular momentum (i.e. no precession). This simplification, while enabling GPU execution, will lead to some differences in posterior results compared to analyses using more comprehensive models, a point I discuss further in Section 4.1.

For the BNS merger GW170817, the discovery paper by Abbott et al. (2017a) uses the IMRPhenomPv2_NRTidal waveform. This model extends IMRPhenomD to include spin precession (Pv2) and, crucially, tidal deformability effects (NRTidal) which are required for BNS mergers like GW170817. While **Ripple** does not support this exact waveform, it does provide the IMRPhenomD_NRTidalv2 model. This model is based on the aligned-spin IMRPhenomD model, but includes tidal effects. Although this **Ripple** model has only been validated for the low spin regime (dimensionless spin components $\chi_{1z}, \chi_{2z} < 0.05$), analyses of GW170817 indicate that precession effects are not dominant for this event, given its low effective precession parameter (Abbott et al., 2017b). This makes IMRPhenomD_NRTidalv2 a suitable waveform for analysis of GW170817, with the aligned-spin assumption expected to introduce only minor differences compared to precessing models for this event. This is discussed in Section 4.2.

3.1.2 Analysis Models and Priors

To systematically test my GPU pipeline, validate its accuracy, and explore the physics relevant to H_0 inference, I defined and ran several distinct analysis configurations, which I refer to as models. These are summarised in Table 1. The key distinctions between these models involved:

- The target event (GW150914 for validation vs. GW170817 for H_0 inference).
- The specific set of parameters included in the estimation (i.e., whether H_0 and v_p were sampled).
- Whether heterodyning (as described in Section 2.7.1) was used to accelerate likelihood calculations.
- The waveform model employed (as discussed in Section 3.1.1).
- The prior distributions and constraints applied to the parameters.

In Table 1, ‘Constrained’ priors refer to relatively narrow priors that are chosen to be centred on, or informed by, previous analysis results for the specific event (e.g., from previous LVK publications). This approach is often used to speed up convergence or to focus the analysis on a particular region of the parameter space. ‘General’ priors refer to broader, astrophysically motivated priors that span a wider range of plausible parameter values, often chosen to be uninformative over that range. Throughout my project, I utilised prior definitions equivalent to those used in the comparative analyses (e.g., LVK results or the study by Wong et al. (2023)) to ensure valid comparisons. For example, the LVK analysis of GW150914 used general priors for all parameters, whilst the H_0 analysis of GW170817 adopted tightly constrained priors (Abbott et al., 2017b).

Table 1: Summary of parameter estimation models I used in this project. ‘ N_{params} ’ refers to the number of sampled parameters. ‘ H_0 Analysis’ indicates whether the Hubble constant (H_0) and peculiar velocity (v_p) were included as sampling parameters.

Model	Event	Priors	N_{params}	Heterodyned	H_0 Analysis	Waveform Model
Bilby	GW170817	Constrained	17	✗	✗	IMRPhenomPv2_NRTidal
A	GW170817	Constrained	15	✓	✓	IMRPhenomD_NRTidalv2
B	GW170817	Constrained	13	✓	✗	IMRPhenomD_NRTidalv2
C	GW170817	Constrained	13	✗	✗	IMRPhenomD_NRTidalv2
D	GW170817	General	15	✓	✓	IMRPhenomD_NRTidalv2
E	GW170817	General	13	✓	✗	IMRPhenomD_NRTidalv2
F	GW150914	General	11	✓	✗	IMRPhenomD
G	GW150914	General	11	✗	✗	IMRPhenomD
A (Flat in z)	GW170817	Identical to A, but with a flat-in-redshift prior instead of a volumetric d_L prior				
A (Uncertainty)	GW170817	Identical to A, but with peculiar velocity uncertainty $\sigma_{v_p} = 250 \text{ km s}^{-1}$ instead of $\sigma_{v_p} = 150 \text{ km s}^{-1}$				

3.2 Parameter Estimation and H_0 Implementation

A key scientific application of my accelerated pipeline was the robust implementation of joint GW source parameter and Hubble constant (H_0) estimation. This section details the methodology I used, drawing from the framework established by [Abbott et al. \(2017b\)](#) for their analysis of GW170817, and showcases an alternate implementation approach facilitated by my pipeline.

3.2.1 Theoretical Framework for H_0 Inference

As outlined in Section 2.3.3, for a nearby bright siren like GW170817, H_0 can be estimated. Following the approach of [Abbott et al. \(2017b\)](#), I assumed statistical independence between the GW data x_{GW} , the host galaxy's recessional velocity v_r , and the estimate of its peculiar velocity $\langle v_p \rangle$. It was assumed that all galaxies in the group associated with NGC 4993 are approximately at the same distance, and thus share the same Hubble flow velocity; this assumption is accurate to within $\sim 1\%$, given the group radius of ~ 0.4 Mpc ([Abbott et al., 2017b](#)).

The combined likelihood for the parameters relevant to H_0 inference ($H_0, d_L, \cos \iota, v_p$) can be written as a product of likelihoods for the different pieces of information:

$$\mathcal{P}(x_{\text{GW}}, v_r, \langle v_p \rangle | d_L, v_p, H_0) = \mathcal{P}(x_{\text{GW}} | d_L, \cos \iota) \mathcal{P}(v_r | d_L, v_p, H_0) \mathcal{P}(\langle v_p \rangle | v_p) \quad (5)$$

The first term, $\mathcal{P}(x_{\text{GW}} | d_L, \cos \iota)$, is the standard GW likelihood of observing the strain data x_{GW} , given a waveform and a set of parameters. When inferring H_0 , the dependence on all GW parameters $\vec{\lambda}$ not directly involved in the H_0 calculation have been marginalised over:

$$\mathcal{L}_{x_{\text{GW}}} = \mathcal{P}(x_{\text{GW}} | d_L, \cos \iota) = \int \mathcal{P}(x_{\text{GW}} | d_L, \cos \iota, \vec{\lambda}) \pi(\vec{\lambda}) d\vec{\lambda} \quad (6)$$

The second term, $\mathcal{P}(v_r | d_L, v_p, H_0)$, is the likelihood of the measured host galaxy's recessional velocity v_r . Assuming a Gaussian uncertainty σ_{v_r} on the measurement of v_r , and taking Equation 2 ($v_r \approx H_0 d_L + v_p$), this term is modelled as:

$$\mathcal{L}_{v_r} = \mathcal{P}(v_r | d_L, v_p, H_0) = \mathcal{N}[v_p + H_0 d_L, \sigma_{v_r}^2](v_r) \quad (7)$$

where $\mathcal{N}[\mu, \sigma^2](x)$ is the Gaussian probability density with mean μ and variance σ^2 , evaluated at x . For NGC 4993, the recessional velocity in the CMB frame is $v_r = 3327 \pm 72 \text{ km s}^{-1}$. This value is taken from the two micron all sky survey (2MASS) ([Crook et al., 2007, 2008](#)), corrected to the CMB frame using measurements by [Hinshaw et al. \(2009\)](#).

The third term, $\mathcal{P}(\langle v_p \rangle | v_p)$, is the likelihood of the peculiar velocity measurement $\langle v_p \rangle$, given the sampled peculiar velocity v_p . This term accounts for the uncertainty in peculiar velocity models. Assuming the estimate $\langle v_p \rangle$ is also Gaussian we have:

$$\mathcal{L}_{\langle v_p \rangle} = \mathcal{P}(\langle v_p \rangle | v_p) = \mathcal{N}[v_p, \sigma_{v_p}^2](\langle v_p \rangle) \quad (8)$$

with $\langle v_p \rangle = 310 \pm 150 \text{ km s}^{-1}$, based on the work of [Carrick et al. \(2015\)](#).

The total log-likelihood I used in the Bayesian inference for the joint estimation of all parameters is then the sum of the standard GW log-likelihood, and the log-likelihoods from these two velocity terms:

$$\ln \mathcal{L}_{\text{total}} = \ln \mathcal{L}_{x_{\text{GW}}} + \ln \mathcal{L}_{v_r} + \ln \mathcal{L}_{\langle v_p \rangle} \quad (9)$$

The full posterior probability for all parameters is then:

$$\mathcal{P}(H_0, d_L, \cos \iota, v_p | x_{\text{GW}}, v_r, \langle v_p \rangle) \propto \mathcal{L}_{x_{\text{GW}}} \mathcal{L}_{v_r} \mathcal{L}_{\langle v_p \rangle} \pi(d_L) \pi(\cos \iota) \pi(v_p) \pi(H_0) \times [\mathcal{N}_s(H_0)]^{-1} \quad (10)$$

The term $\mathcal{N}_s(H_0)$ accounts for the selection effects ([Abbott et al., 2016b](#)), related to the probability of detecting a GW event. A detailed discussion of this is beyond the scope of my project, however the

analysis in [Abbott et al. \(2017b\)](#) concluded that if priors are set on distances rather than redshifts, this selection term is approximately constant and can be ignored.

The priors $\pi(\cdot)$ used in my H_0 analysis, following [Abbott et al. \(2017b\)](#), were:

- Luminosity distance d_L : A volumetric prior, $\pi(d_L) \propto d_L^2$ for $d_L \in [10, 75]$ Mpc. This reflects a uniform distribution in Euclidean space.
- Hubble constant H_0 : A flat-in-log prior, $\pi(H_0) \propto 1/H_0$ for $H_0 \in [45, 250]$ km s⁻¹ Mpc⁻¹.
- Peculiar velocity v_p : A flat (uniform) prior, $\pi(v_p) \propto 1$, for $v_p \in [-1000, 1000]$ km s⁻¹.
- Binary inclination angle $\cos \iota$: A flat (uniform) prior, $\pi(\cos \iota) \propto 1$ for $\cos \iota \in [-1, 1]$. This corresponds to an isotropic distribution of orientations.
- Other GW parameters $\vec{\lambda}$: These used standard priors commonly employed in LVK analyses of GW170817 ([Abbott et al., 2017b](#); [Wong et al., 2023](#)).

3.2.2 Inference Implementation

The exact implementation methods used in the LVK paper [Abbott et al. \(2017b\)](#) to marginalise (integrate) over the GW parameters (Equation 6) and combine the three likelihood terms (Equation 9) is not fully detailed in their paper. Discussions with other computational astrophysicists working on gravitational-wave Bayesian inference also did not yield a definitive clarification of the precise step-by-step methodology used by the authors. However, it appears they have performed the calculation in a two-step process: first obtaining the posterior samples, and then performing post-processing using the likelihood terms to infer H_0 .

In contrast, I implemented a more direct, one-step approach using my JAX-based pipeline. I directly incorporated the Hubble constant H_0 and peculiar velocity v_p as sampling parameters within the NS run, alongside all other GW source parameters $\vec{\lambda}$. I defined prior functions for these new parameters (H_0, v_p) as described above. The standard GW log-likelihood function was then augmented by adding the two velocity log-likelihood terms ($\ln \mathcal{L}_{v_r}$, and $\ln \mathcal{L}_{\langle v_r \rangle}$) to form the total log-likelihood (as in Equation 9). The Gaussian probability densities required for these terms were evaluated using standard functions from the SciPy library.

This direct sampling approach was facilitated by the flexibility and modular design of my JAX pipeline. Adding the additional parameters, priors, and likelihood required the modifications of only a few lines of code. This contrasts with more involved modifications or post-processing steps if this were to be implemented in other frameworks like Bilby.

3.2.3 Prior Sensitivity

[Abbott et al. \(2017b\)](#) investigated the sensitivity of their H_0 result to two alternative modelling assumptions:

- (a) Altering their prior choice for distance, specifically comparing their default volumetric prior $\pi(d_L) \propto d_L^2$ to an alternative prior that is flat-in-redshift, $\pi(z) = \text{const}$.
- (b) Assuming a greater uncertainty for the peculiar velocity correction of NGC 4993, specifically $\sigma_{v_p} = 250$ km s⁻¹ instead of the canonical value of 150 km s⁻¹.

The paper concluded that the impact on the H_0 result for both of these alternative models was small compared to the overall statistical uncertainty of their measurement. Their H_0 results changed by less than 1σ , as measured by statistical error of their canonical analysis. However, as I will show in Section 4.3.1 I dispute their conclusion regarding the flat-in-z prior based on my re-analysis.

The rationale for testing an increased peculiar velocity uncertainty was to assess assumptions that NGC is a member of the nearby group of galaxies, and that its group has centre of mass velocity close to Hubble flow. The flat-in-redshift prior explores the impact of assuming a different underlying source

distribution. This is motivated by the idea of first making a redshift measurement for the host galaxy, and then using that redshift as a prior when analysing the GW data. This alternative assumption tends to up-weight nearby (low- z , low- d_L) sources, compared to the canonical volumetric prior. This tests the robustness of the H_0 inference against assumptions about the underlying population of GW sources.

The authors of [Abbott et al. \(2017b\)](#) implemented the flat-in- z prior by reweighting the posterior samples obtained from their main run, which used the volumetric prior. The reweighting formula involves the ratio of the priors and the Jacobian of the transformation $d_L(z)$. This would be:

$$\begin{aligned}\pi_{\text{new}}(z) &= \frac{1}{z_{\text{max}} - z_{\text{min}}} \quad (\text{uniform prior in } z) \\ \pi_{\text{old}}(d_L) &= \frac{3d_L^2}{d_{L,\text{max}}^3 - d_{L,\text{min}}^3} \quad (\text{volumetric prior in } d_L)\end{aligned}$$

Equation 3 gives us the Jacobian for the transformation: $|dz/dd_L| \approx H_0/c$. Thus the reweighting factor w_i^* for a sample i with original weight w_i is:

$$w_i^* = w_i \cdot \frac{\pi_{\text{new}}(z)}{\pi_{\text{old}}(d_L(z)) \cdot \left| \frac{dd_L}{dz} \right|} = w_i \cdot \frac{\pi_{\text{new}}(z)}{\pi_{\text{old}}(d_L(z))} \cdot \left| \frac{dz}{dd_L} \right| \quad (11)$$

A naive implementation of this reweighting approach can be problematic, particularly if the original samples (π_{old}) do not adequately cover regions of parameter space that are strongly favoured by the new prior (π_{new}). I discuss this further in Section 4.3.1.

My JAX pipeline allowed me to easily perform the analysis for the flat-in- z prior by directly sampling with this prior. This involved defining a new prior function that samples z uniformly within bounds corresponding to the original d_L range, for a fiducial H_0 . The likelihood evaluation then requires converting this sampled z back to d_L via a cosmological conversion, which required refactoring of standard libraries such as **Astropy**.

3.3 GPU-Specific Optimisations and Framework

The overarching goal driving my methodological choices was to create a fast, flexible, and understandable Bayesian inference pipeline capable of handling GW data analysis efficiently using GPUs. The pipeline needed to be sufficiently rapid for timely analysis, and I aimed for a modular design to facilitate easy extensibility for future work, such as incorporating new waveform models or additional physical effects such as wrap-around parameters.

3.3.1 The JAX Framework

GPUs were chosen as the target hardware due to their inherent advantages for the computational workload of GW parameter estimation, as outlined in Section 2.7.2. For the software framework, I selected JAX ([Bradbury et al., 2018](#)), a Python library for high-performance numerical computing, for several key reasons:

- JAX is designed for accelerating numerical computation on GPUs with minimal code changes compared to standard NumPy usage, facilitating straightforward code development. It uses the Accelerated Linear Algebra (XLA) compiler to optimise and fuse computations, generating highly efficient machine code. I made extensive use of the `jax.jit` (just-in-time) decorator to enable these XLA optimisations in my pipeline.
- JAX integrates seamlessly with a growing ecosystem of numerical and scientific libraries, including Bayesian inference libraries like BlackJAX ([Cabezas et al., 2024](#)), and GW-specific libraries such as Jim ([Wong et al., 2023](#)), and Ripple ([Edwards et al., 2023](#)). This existing ecosystem provided crucial building blocks for my pipeline.

- JAX’s powerful function transformations such as `jit`, `vmap` (automatic vectorisation/batching), and `pmap` (parallelisation across multiple devices), are crucial for achieving high performance. I utilised `vmap` to efficiently batch likelihood evaluations across all live points, enabling order of magnitude acceleration in likelihood evaluation.
- While not the primary focus for the NS implementation used here, JAX’s robust automatic differentiation capabilities are valuable for potential future extensions using gradient-based samplers like HMC.

The specific nested sampling algorithm employed in this project is the nested slice sampling algorithm implemented by (Yallup et al., 2025). This is a novel vectorised nested sampler which leverages GPU acceleration through the JAX framework, allowing the entire inference process (prior evaluation, likelihood evaluation, sampling steps) to run efficiently on GPUs. It makes extensive usage of `jax.jit` compilation.

Using JAX effectively requires careful code design. Functions intended for `jit` compilation must be ‘pure’ (i.e. self-contained) and work with traceable array objects. This necessitates a different way of thinking about code structure compared to standard Python. A completely `jit` compiled pipeline would require extensive refactoring of current GPU parameter estimation libraries. My focus was on `jit`-compiling the most computationally intensive parts, namely the likelihood evaluations and core nested sampling loop, which I successfully achieved.

3.3.2 The Jim Framework and Adaptations

The Jim library, developed by (Wong et al., 2023), is a JAX-based GW inference library that is currently under heavy development. My initial exploration of the library revealed that some existing example implementations had not yet been fully validated. During this process, I also uncovered an issue related to the loading function for PSD data from GWOSC within Jim and contributed a fix⁶ for it. Due to these minor inconsistencies and the library’s rapidly evolving developmental stage, I choose not to directly utilise the full Jim library.

Instead, a significant part of my work involved adapting and distilling essential components from the Jim library, with guidance from my supervisor W.J. Handley. The approach I adopted was to extract the core functionalities needed for my pipeline, and build a more streamlined, modular, and understandable system around them. Specifically:

- I adopted and adapted implementations from Jim for handling detector data, implementing my fixes, (loading strain data, calculating PSDs, defining detector response functions) and the interface to the Ripple library (Edwards et al., 2023) for JAX-based waveform generation. The core structure for evaluating the standard GW likelihood function was also based on Jim’s formulation of the standard GW likelihood.
- I developed custom Python code for prior definitions, parameter handling and transformation, the main script structure for running the nested sampler, and the specific implementation of the combined $\text{GW}+H_0$ likelihood. For example, I replaced complex class structures for priors in Jim with simpler, modular functional definitions. This made it easier to define and modify priors for different parameters, including the new parameters H_0 and v_p .
- I consciously removed all non-essential code, bulky class hierarchies, and unused functionalities to create a ‘bare-bones’ implementation of Jim. My goal was to create a minimal, understandable, and modifiable codebase where the core logic of the parameter estimation process was transparent. This also aided in debugging and ensuring the correctness of my implementation.

This process resulted in a functional JAX-based parameter estimation pipeline that, while building on foundational elements from Jim and Ripple, is largely my own streamlined implementation tailored to the specific goals of this project. It is designed to be easy to understand, modify, and extend. My

⁶<https://github.com/kazewong/jim/pull/196>

framework leverages standard, well-tested libraries like `SciPy` for auxiliary tasks where `JAX` equivalents were not necessary, ensuring numerical accuracy and compatibility, while relying as little as possible on the more complex or rapidly changing aspects of the aforementioned developmental libraries.

3.4 Code Development and Workflow

As mentioned above, the development process focused on creating an extensible, modular, and streamlined pipeline. The overall development process I followed consisted of these key steps:

- Step 1: Initial Setup and Refactoring** I began with a `Jim` example script for GW150914 parameter estimation, which was provided in the `Jim` documentation. Under the guidance of my supervisor, W.J. Handley, I systematically stripped out unnecessary `Jim` abstractions, replacing them with direct calls to the `BlackJAX` nested sampler and custom functions I wrote for managing priors and parameters transformations. I achieved maximum clarity and minimal dependencies beyond `JAX`, `BlackJAX`, and `Ripple`. This strategy facilitated easier debugging, and ensured a robust implementation, which was particularly important given `Jim` has no stable releases. This careful approach also led to the identification and subsequent fixing of the PSD loading issue in `Jim`.
- Step 2: Validation with GW150914** I thoroughly tested my pipeline, including both heterodyned and non-heterodyned versions, on the benchmark event GW150914. I compared the posterior results I obtained against standard LVK posteriors from GWOSC, as well as `Bilby` outputs from other papers. This step was crucial to ensure that the core parameter estimation machinery was functioning correctly.
- Step 3: Extension to GW170817** Once confident with the GW150914 results, I adapted the pipeline for the BNS event GW170817. This involved incorporating the longer data duration, adjusting frequency ranges for analysis, and utilising the appropriate BNS waveform model (including tidal effects). Again, I validated my results against `Bilby` outputs and LVK posteriors, carefully considering differences that might arise from model choices. During this phase, I also spent considerable time refactoring my code for optimisation and organisation, deepening my understanding of `JAX` constructs. However, my primary focus throughout remained on the accuracy of implementation.
- Step 4: H_0 Implementation** With a validated pipeline for parameter estimation, I then implemented the H_0 inference capability outlined in Section 3.2. This involved modifying the likelihood function to include the two extra terms, and adding H_0 and the peculiar velocity v_p as sampling parameters, with their respective priors.
- Step 5: Comparative Analysis Runs and Prior Sensitivity** Finally, I performed a series of production runs using the different analysis models defined in Table 1 to assess the performance and scientific output of my pipeline. I compared the runtime performance of my GPU pipeline against CPU-based `Bilby` runs, using equivalent setups where possible. I also conducted the specific runs required to investigate the prior sensitivity for the H_0 measurement, implemented in Abbott et al. (2017b).

The resulting code features a modular structure. This makes it straightforward to modify priors, add or remove parameters, or change waveform models by editing specific, well-defined functions or configuration lists within my code. This modularity was key to easily extending the pipeline for H_0 inference, and conducting the prior sensitivity studies.

4 Results and Discussion

This section presents the main results of my project. I begin by validating the scientific accuracy of my GPU pipeline by comparing its results for the BBH merger GW150914 against standard LVK results (Section 4.1). The core scientific application of my work follows, with the analysis of the bright siren GW170817 (Section 4.2) and inference of the Hubble constant. This includes a critical discussion of

prior sensitivity (Section 4.3). I then quantify the performance difference between the developed GPU pipeline and traditional CPU methods (Section 4.4). Finally, I discuss the limitations of the current work (Section 4.5).

Before presenting the corner plots, I define the key parameters commonly displayed:

- Chirp Mass $M_c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$, where m_1 and m_2 are the component masses. M_c is the best-constrained mass parameter from the inspiral phase of the GW signal.
- Mass ratio $q = m_2 / m_1$, where $m_1 \geq m_2$, thus $0 < q \leq 1$. This quantifies the asymmetry in the masses of the two components.
- Luminosity distance d_L is the distance to the source as inferred from the GW signal amplitude.
- Inclination angle ι is the angle between the binary’s orbital angular momentum vector and the line of sight to the observer. $\iota = 0^\circ$ is face-on. $\iota = 90^\circ$ is edge-on. This parameter is degenerate with d_L in its effect on the observed waveform amplitude.

Other parameters such as spin magnitudes are also inferred, but are not shown in summary plots for brevity unless particularly relevant to the discussion. I focus on the above parameters as they are key for H_0 inference.

4.1 Validation with GW150914

To ensure that my GPU-accelerated pipeline produces scientifically accurate results, I first validated it against the well-studied benchmark BBH event GW150914. I performed parameter estimation using my pipeline (Models F and G from Table 1) and compared the resulting posterior distributions to those released by the LVK collaboration (Abbott et al., 2019b), and also to results generated by Wong et al. (2023) using Bilby with the same IMRPhenomD waveform. I used identical prior distributions and the same publicly available GWOSC strain data for these comparisons.

As discussed in Section 3.1.1, the LVK posterior samples for GW150914, taken from the GWTC-2.1 catalogue (Collaboration et al., 2022) (specifically the ‘no_cosmo’ release⁷), were generated using the IMRPhenomXPHM waveform model. This model includes spin precession and higher-order modes, whereas my JAX pipeline, and the Bilby comparison from Wong et al. (2023) used IMRPhenomD, which assumes aligned-spins, and only the dominant quadrupole radiation. Previous studies on GW150914 (Abbott et al., 2016a) indicate that while differences in posteriors can arise from these waveform model choices, IMRPhenomD should still capture the dominant features of the main source parameters for a first-level validation, potentially with slight shifts in the peaks or widths of some parameter posteriors.

Figure 3 provides an internal consistency check for my pipeline, and a comparison with a Bilby run using the same simplified waveform. It shows posterior distributions from my heterodyned JAX (Model F, purple), without heterodyning (Model G, yellow), and against a Bilby run from Wong et al. (2023) (green).

⁷More information on the data release: <https://gwosc.org/GWTC-2.1>

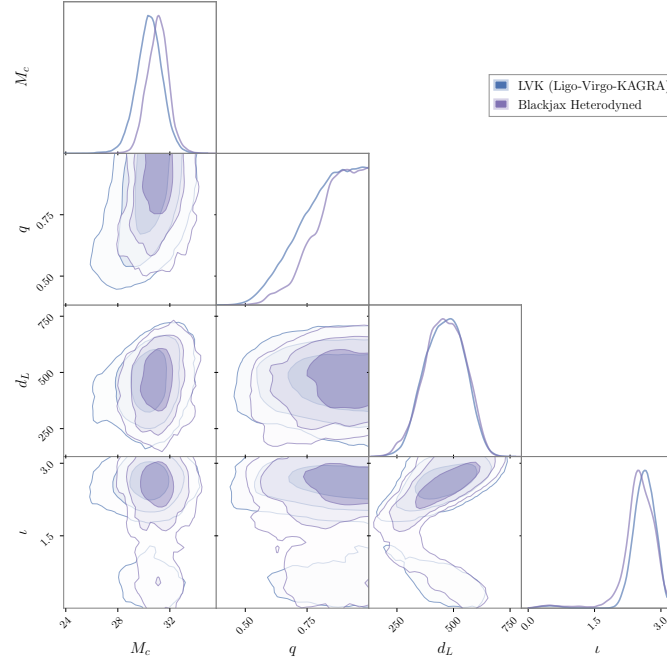


Figure 2: Corner plot comparing posterior distributions for GW150914. Key parameters shown are chirp mass (M_c), mass ratio (q), luminosity distance (d_L), and inclination angle (ι). LVK results shown in blue. JAX pipeline Model F (heterodyned) shown in purple.

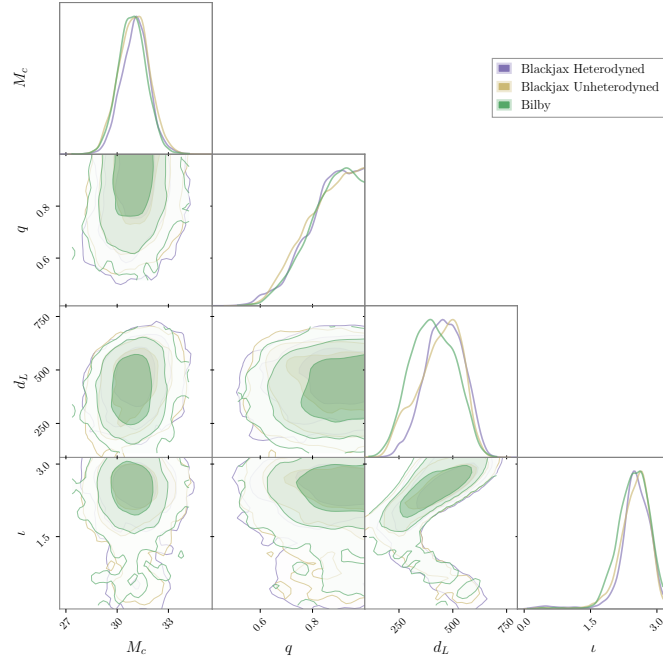


Figure 3: Corner plot comparing posterior distributions for GW150914. This is an internal consistency check for my pipeline. Key parameters shown are (M_c , q , d_L , and ι). Model F (heterodyned) is shown in purple. Model G (unheterodyned) is shown in yellow. Bilby run from [Wong et al. \(2023\)](#) is shown in green.

Observing Figure 2, my results (purple) show broad agreement with the LVK samples (blue). There is a noticeable discrepancy in the chirp mass M_c , where my pipeline’s posterior for M_c is slightly shifted towards higher values compared to the LVK samples. These discrepancies are likely primarily due to the difference in waveform models. These minor differences are expected when considering the differing waveform complexities.

More importantly, Figure 3 demonstrates excellent internal consistency for my pipeline and strong agreement with an independent **Bilby** analysis that used the same **IMRPhenomD** waveform. My heterodyned (Model F, purple) and unheterodyned (Model G, yellow) **JAX** runs produce very similar posterior distributions. Furthermore these results show considerable overlap and good agreement with the **Bilby** posterior samples from [Wong et al. \(2023\)](#). While there is an inconsistency in the d_L posteriors, Figure 2 shows that the posterior samples produced by my pipeline are identical to the LVK sample. This consistency across different implementations using the same underlying model is a crucial validation step. It confirms that the core Bayesian inference engine (the **BlackJAX** nested sampler I implemented) and my likelihood implementation are functioning correctly and are consistent with established CPU-based tools.

Overall, the recovery of source parameters that are generally consistent with standard methods and official results, within the expectations set by the chosen waveform model, provides confidence in my pipeline’s fundamental correctness. This successful validation was a prerequisite for applying my pipeline to more complex analyses, such as the H_0 inference from GW170817.

4.2 Bright Siren Analysis: GW170817

Having validated my pipeline with GW150914, this section details my analysis of the BNS merger GW170817. This analysis served as a precursor to, and an integral part of, the H_0 inference detailed in Section 4.3. As discussed in Section 3.1.1, I employed the **IMRPhenomD_NRTidalv2** waveform model.

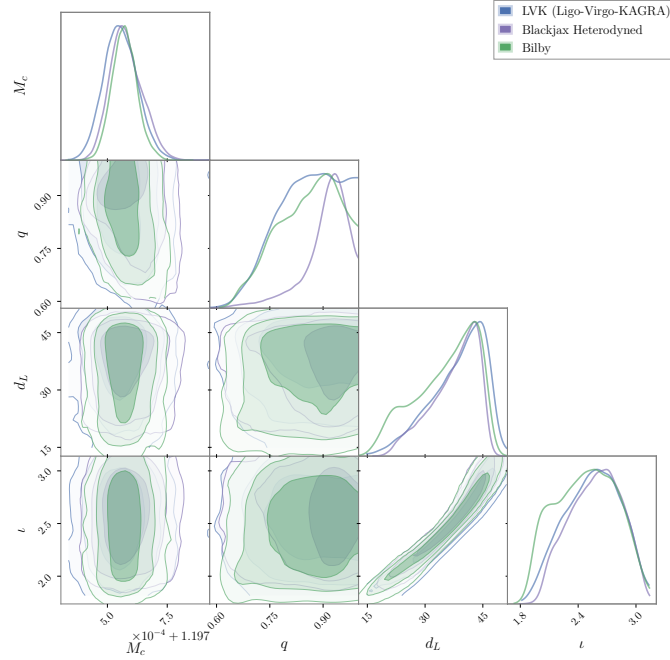


Figure 4: Corner plot comparing posterior distributions for GW170817. Key parameters shown are (M_c , q , d_L , and ι). LVK (**IMRPhenomPv2_NRTidal** waveform) results are shown in blue. **JAX** Model A (heterodyned, H_0 , **IMRPhenomD_NRTidalv2** waveform) shown in purple. **Bilby** run is shown in green (Model **Bilby**, **IMRPhenomPv2_NRTidal** waveform).

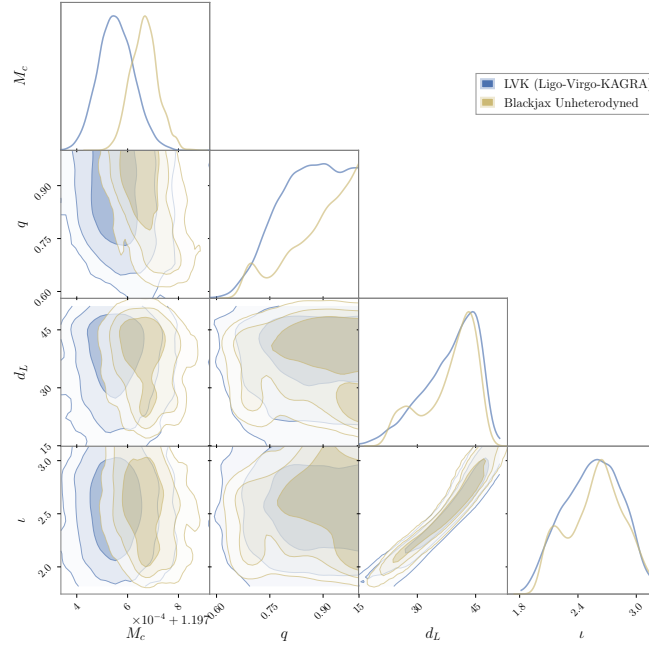


Figure 5: Corner plot comparing posterior distributions for GW170817. Key parameters shown are (M_c , q , d_L , and ι). LVK (IMRPhenomPv2_NRTidal waveform) results are shown in blue. JAX Model C (unheterodyned, IMRPhenomD_NRTidalv2 waveform) shown in yellow.

In Figure 4, my heterodyned H_0 inference run (Model A, purple) shows good agreement with my Bilby reference run (green) and the LVK results (blue) for M_c , d_L , and ι . The mass ratio q posterior from my Model A run (maximum a posteriori, MAP, $q = 0.93^{+0.05}_{-0.05}$) peaks sharply at a high value, and has lower posterior density at lower values of q compared to both the LVK and Bilby results, which find q broadly distributed in the range 0.73 – 1.

The unheterodyned JAX run (Model C, yellow) in Figure 5 also shows a chirp mass peak slightly shifted to higher values compared to the LVK sample (blue). Its mass ratio similarly peaks at a higher value of q , and has lower posterior density at lower q values. However, the two JAX models do not have agreeing posterior distributions for their mass ratios.

The differences in chirp mass and mass ratio posteriors, particularly the tendency for my JAX runs to prefer higher q values, might be attributable to several factors. The primary suspect is the difference in waveform models, as unlike the GW150914 analysis, the Bilby samples in this case used the same waveform as the standard LVK samples. Although precession effects in this case are considered small, they could still subtly influence parameter correlations and posteriors. Another factor is the lack of ‘wrap-around’ parameters in my JAX code such as phase or sky angles, which are periodic. This is currently being investigated as a source of error by members of the Handley Research Group⁸.

However, the recovered mass ratio is still enclosed in the broader range of the LVK and Bilby posteriors, and is also within the broader range reported in the literature (Abbott et al., 2019a; Narikawa et al., 2020). Despite these minor differences, the overall consistency in recovering the key parameters, especially d_L and ι , which are crucial for H_0 inference, gave me confidence to proceed with the Hubble constant estimation.

⁸More detail can be found at: <https://handley-lab.co.uk>

4.3 An Estimate of H_0 from GW170817

Using the methodology I outlined in Section 3.2, I performed a joint inference of the GW source parameters and the Hubble constant H_0 (along with the peculiar velocity v_p of the host galaxy NGC 4993). This section presents the results of my analysis, comparing them to the original findings of Abbott et al. (2017b) and then investigating the impact of prior assumptions on H_0 inference.

To ensure that the process of jointly fitting for H_0 and v_p did not perturb the inference of the primary GW source parameters, I first compared runs performed with and without the H_0 fitting, keeping all other settings identical (i.e., comparing Model A, which includes H_0 and v_p , with Model B). Figure 6 overlays the posterior distributions for the main astrophysical parameters from Model A (purple) and Model B (yellow), with the standard LVK posterior samples (blue) for reference.

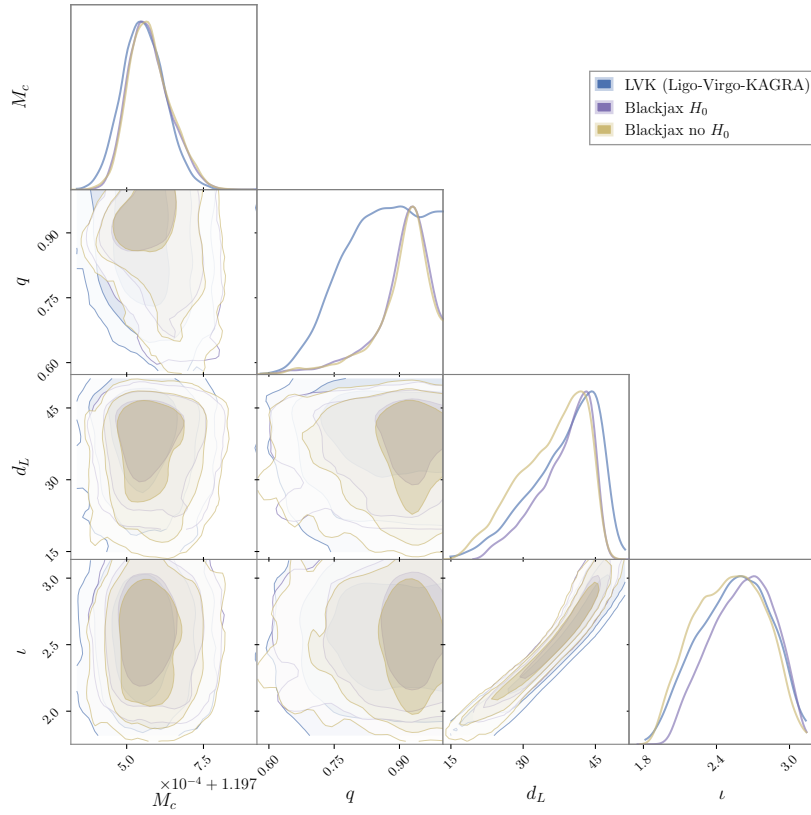


Figure 6: Corner plot comparing posterior distributions for GW170817. Key parameters shown are (M_c , q , d_L , and l). LVK results shown in blue. JAX pipeline Model A (H_0 , v_p) shown in purple. JAX pipeline Model B (no H_0 , v_p) shown in yellow.

As evident from Figure 6, the posterior distributions for the GW source parameters from my JAX runs are nearly identical, whether H_0 and v_p are sampled or not. This is an important consistency check, confirming that the additional likelihood terms primarily constrain H_0 and v_p themselves, and do not introduce significant changes to the inference of core GW parameters.

Figure 7 shows the marginalised posterior probability distribution for H_0 that I obtained from my JAX pipeline (Model A). This result is directly compared with the result published by Abbott et al. (2017b). For context, constraints from early-universe CMB measurements by Planck (Ade et al., 2016) and late-universe SNe measurements by SH0ES (Riess et al., 2016) are also shown.

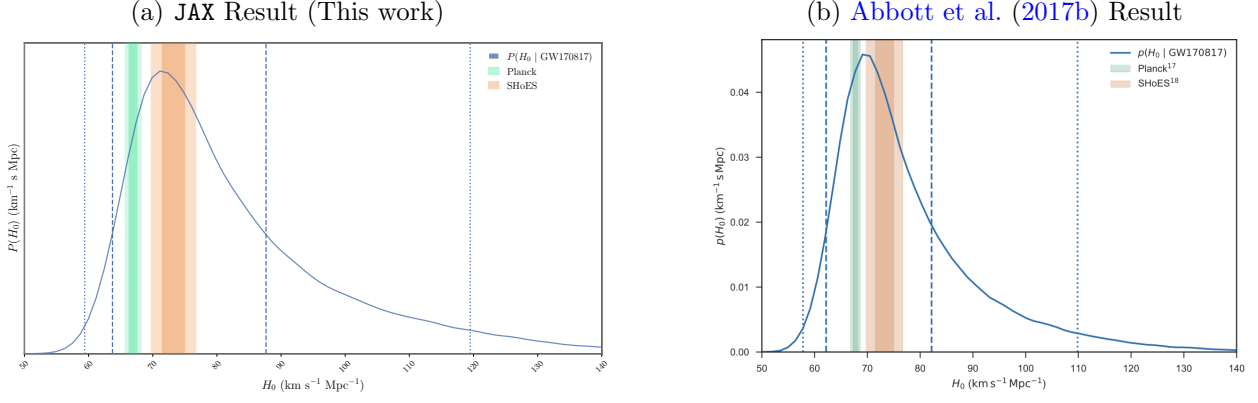


Figure 7: Comparison of the marginalised posterior density for H_0 from GW170817. Constraints from Planck (Ade et al., 2016) (green shaded region) and SHoES (Riess et al., 2016) (orange shaded region) are shown for context. The 1σ and 2σ credible intervals are indicated by dashed and dotted lines respectively. The blue curves show the posterior density. (a) Result I obtained from my JAX pipeline (Model A) with the MAP value and minimal 68.3% credible interval found to be $H_{0,\text{JAX}} = 70.4^{+14.9}_{-6.5} \text{ km s}^{-1} \text{ Mpc}^{-1}$. (b) Original result from Abbott et al. (2017b), who reported $H_{0,\text{LVK}} = 70.0^{+12.0}_{-8.0} \text{ km s}^{-1} \text{ Mpc}^{-1}$.

My result for the Hubble constant, obtained from Model A, is $H_{0,\text{JAX}} = 70.4^{+14.9}_{-6.5} \text{ km s}^{-1} \text{ Mpc}^{-1}$. This is in excellent agreement with the LVK result of $H_{0,\text{LVK}} = 70.0^{+12.0}_{-8.0} \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Abbott et al., 2017b). The MAP values of the distributions are very consistent, and the 68.3% credible intervals overlap significantly. The slight differences in the width and asymmetry of the credible intervals are within expectations arising from minor variations in sampler behaviour and waveform models (IMRPhenomD_NRTidalv2 vs. IMRPhenomPv2_NRTidal). This successful reproduction of the landmark H_0 measurement from GW170817 using my JAX pipeline is a key validation of my methodology and demonstrates its capability for cosmological inference.

4.3.1 Prior Sensitivity Analysis

Having reproduced the baseline H_0 result, I then re-examined the discussion on prior sensitivities presented in Abbott et al. (2017b) and laid out in Section 3.2.3. Specifically, I investigated the impact of: (a) using a flat-in-redshift prior instead of a volumetric d_L prior, and (b) increasing the uncertainty on the peculiar velocity estimate $\sigma_{\langle v_p \rangle}$ from 150 km s^{-1} to 250 km s^{-1} .

First, to ensure I understood their method, I replicated the reweighting approach used by Abbott et al. (2017b) for the flat-in-z prior. I took the posterior samples from my baseline run (Model A, whose H_0 posterior is shown in Figure 7a), and applied the reweighting factor outlined in Section 3.2.3. The results of this reweighting are shown in Figure 8a, compared directly to the corresponding figure from Abbott et al. (2017b) in 8b.

As seen in Figure 8, my reweighted flat-in-z curve (blue in panel 8a) closely matches the corresponding blue curve in panel 8b from Abbott et al. (2017b). This suggests I have correctly reproduced their reweighting methodology. Both reweighted curves show only a minor shift in the H_0 posterior compared to the baseline black curve, leading Abbott et al. (2017b) to conclude that the choice of this distance/redshift prior has little impact on the H_0 measurement.

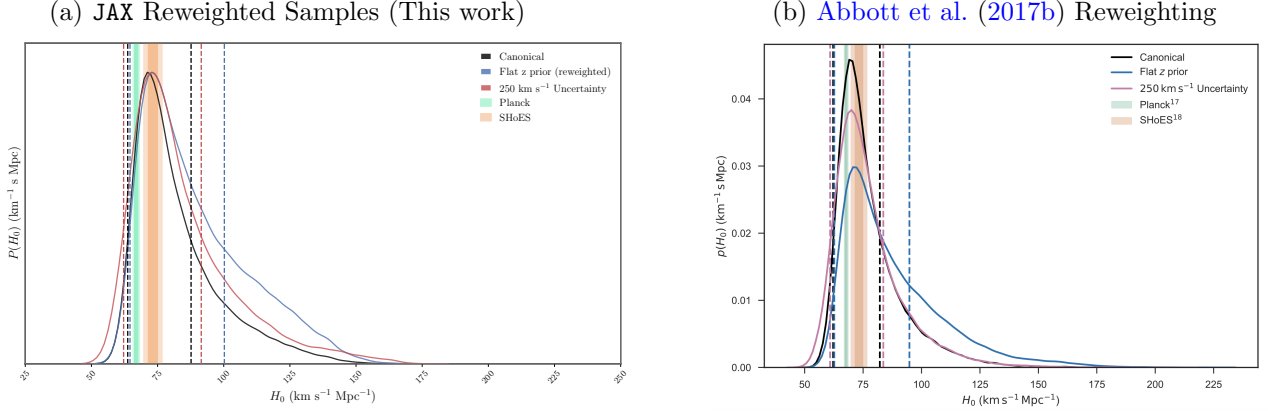


Figure 8: Comparison of H_0 prior sensitivity results. In both panels, the main posterior distribution (volumetric d_L prior, $\sigma_{\langle v_p \rangle} = 150 \text{ km s}^{-1}$) is shown in black. The posterior resulting from reweighting to a flat-in- z prior is shown in blue. The posterior resulting from an increased peculiar velocity uncertainty ($\sigma_{\langle v_p \rangle} = 250 \text{ km s}^{-1}$) is shown in red. Minimal 1σ credible intervals are shown by dashed lines. Constraints for Planck and SHoES are shown as in Figure 7. (a) My result, obtained by reweighting samples from my JAX run (Model A). (b) Original figure from Abbott et al. (2017b).

However, I argue that this reweighted posterior distribution for the flat-in- z prior is incorrect, or at least highly misleading, due to the inherent limitations of posterior reweighting. As discussed in Section 3.2.3, the flat-in-redshift prior up-weights samples with lower d_L values, and reweighting can only redistribute probability mass among the existing samples from the original run. Figures 7a and 7b both show a low number of samples at high H_0 values, and a corresponding low number of samples at low d_L values in Figure 5 (blue and purple curves). Thus, the reweighting procedure assigns small effective weights to these areas due to the scarcity of original samples there, artificially constraining the samples.

To investigate this critical point, I used my JAX pipeline to perform a new nested sampling run (Model A (Flat-in- z) from Table 1), where I directly sampled using a flat-in-redshift prior. The results of this are shown in Figure 9.

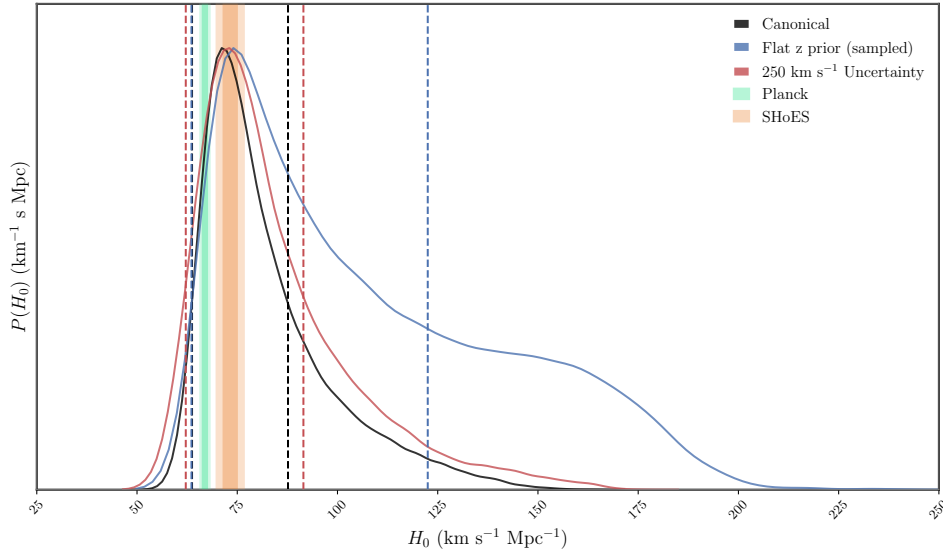


Figure 9: H_0 prior sensitivity analysis using direct sampling. The main posterior distribution is shown in black. The posterior resulting from resampling with a flat-in- z prior is shown in blue. The posterior resulting from an increased peculiar velocity uncertainty is shown in red. Minimal 1σ credible intervals are shown by dashed lines. Constraints for Planck and SHoES are shown as in Figure 7.

The blue curve in Figure 9, which I obtained from direct sampling with a flat-in-redshift prior, shows a markedly different shape for the H_0 posterior compared to the reweighted blue curves in Figure 8. My direct sampling result (blue) shows a much broader distribution with significantly more probability mass extending to higher values of H_0 . This occurs because the sampler is able to more effectively explore solutions with smaller luminosity distances, and isn't artificially constrained. The posterior distribution also suggests there is a valid model for the GW data at a lower luminosity distance. This is illustrated in Figure 10, which compares the marginalised posterior for d_L and ι from the directly sampled run (blue) and the reweighted samples (purple).

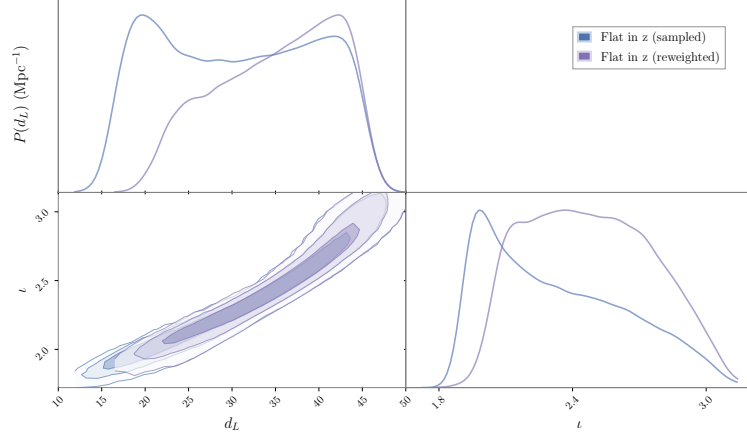


Figure 10: Comparison of the marginalised posterior distribution for luminosity distance d_L and inclination angle ι . Blue: d_L posterior from directly sampling with a flat-in-z prior. Purple: d_L posterior from reweighting the baseline samples.

This finding strongly suggests that the reweighting approach used by Abbott et al. (2017b) was insufficient to capture the true impact of that prior choice due to poor sample coverage in the critical region of parameter space (low d_L). Their conclusion that the flat-in-z prior has a negligible effect appears to be an artifact of the reweighting method's limitations when the initial and target priors differ substantially in the tails or other critical regions of the parameter space. My results suggest that the H_0 inference is quite sensitive to the choice between a volumetric d_L prior and a flat-in-z prior, with the latter allowing for a much broader range of H_0 values and shifting the peak slightly higher. This highlights a key advantage of a fast and flexible inference pipeline: it enables robust sensitivity analyses through direct simulation under various assumptions, rather than relying on post-processing techniques. The red curves in the above figures are consistent across the two implementations, as this does not effect the favoured regions of other parameters.

4.3.2 Bayesian Model Comparison

Nested Sampling directly calculates the Bayesian evidence $\mathcal{Z}$, which allows for quantitative model comparisons. A higher evidence indicates that the model provides a better fit to the data, penalised for unnecessary complexity. I compared the log-evidences for the three models used in my prior sensitivity analysis (Figure 9), in Table 2.

Table 2: Comparison of Bayesian log-evidences ($\ln \mathcal{Z}$) for GW170817 H_0 sensitivity models.

Model Description	$\ln \mathcal{Z} \pm \sigma_{\ln \mathcal{Z}}$
Baseline (volumetric d_L prior, $\sigma_{v_p} = 150$)	638.97 ± 0.07
Increased Uncertainty (volumetric d_L prior, $\sigma_{v_p} = 250$)	639.04 ± 0.06
Flat-in-z Prior (redshift prior, $\sigma_{v_p} = 150$)	639.64 ± 0.06

The log-evidence values presented in Table 2 are very similar for all three models. The differences in

$\ln \mathcal{Z}$ are small, with the Bayes factors $\mathcal{B}_{ij} = \exp(\ln \mathcal{Z}_i - \ln \mathcal{Z}_j)$ being close to 1. The largest Bayes factor, between the flat-in- z prior and the baseline model, is $\mathcal{B} = 1.95$. According to Jeffreys scale, this is ‘barely worth mentioning’. Detailed analysis or data from more bright sirens will be necessary to gain more discriminating power between these underlying assumptions. The slightly higher evidence for the flat-in- z prior is likely attributed to its better fit in the low- d_L or high- H_0 regime.

4.4 CPU vs. GPU Performance

A primary motivation for my project was to accelerate GW parameter estimation using GPUs. As discussed in Sections 2.7.2 and 3.3, GPUs offer theoretical advantages for the parallel computations inherent in nested sampling. Here, I provide quantitative comparisons of runtime and an indicative cost-effectiveness analysis. The parameter estimation runs I performed using **Bilby** (Model Bilby) were run on the University of Cambridge (UoC) CSD3 cluster, primarily using compute nodes equipped with Intel Xeon Ice Lake CPUs. All GPU-based parameter estimation runs using the **JAX** pipeline were performed on a NVIDIA L4 Tensor Core GPU, rented through the Google Cloud Platform⁹.

4.4.1 Cost Analysis

Directly comparing raw runtimes across different hardware architectures can be misleading. A more informative comparison considers the cost or resource allocation required to achieve a given runtime. I use publicly available pricing and institutional charge rates as proxies for relative computational power, assuming a roughly efficient market where price correlates with performance within a given class of hardware.

Internal Cambridge HPC charging rates¹⁰ and DiRAC (Distributed Research utilising Advanced Computing) resource allocation values¹¹ provide one benchmark. Typically, GPU resources are considered significantly more expensive than CPU cores in terms of allocation units or hourly charges. DiRAC and HPC allocations charge 55× more per Wilkes3 GPU hour compared to its Cumulus CPU. Moreover, at the same service level, e.g. Service Level 3, the maximum number of GPU hours available per quarter is 1.5% of the CPU core hours (3000 GPU hours vs. 200,000 CPU core hours), reflecting both hardware cost and scarcity. These figures suggest GPUs are 4.4× more expensive per PetaFlop hour than CPU cores, and are considered $\sim 67\times$ scarcer.

Table 3 provides a snapshot of commercial cloud pricing (Google Cloud, April 2024), comparing CPU instances similar in nature to the Cambridge cluster nodes with various GPU options, including the L4 used in this work. A C2 instance with 60 vCPUs and 4GiB/vCPU serves as a proxy for a typical Ice Lake node (76 cores, 3.3 GiB RAM/core). Prices depend on commitment duration.

Table 3: Indicative CPU vs. GPU Cost Comparison using the Google Cloud Pricing Calculator (as of Apr 2024, prices approximate and subject to change). Costs are shown for a 1-month commitment, 1- and 3-year sustained use discounts.

	Model	Specifications		Price (\$ / Month)		
		vCPUs	RAM / vCPU (GiB)	1 Month	1 Year	3 Years
CPU	C2	60	4	2,288	1,442	916
	C2	480	4	14,591	11,523	7,317
GPU	L4	4	4	520	325	232
	A100	12	7.1	2,682	1,689	939
	H100	26	9	8,075	5,601	3,546

Note: CPU instance includes CPU+RAM cost. GPU prices are for the GPU accelerator only, requiring a small additional cost for the attached CPU/RAM instance.

⁹<https://cloud.google.com>

¹⁰UoC ‘charges for internal users’ page

¹¹<https://docs.hpc.cam.ac.uk/hpc/user-guide/policies.html>

Comparing a single NVIDIA L4 GPU to our proxy for an Ice Lake node, a 60-vCPU C2 node, the GPU appears significantly cheaper on a per-unit basis. As I will show in Section 4.4.2, a single L4 GPU offers a $\sim 30\times$ speed-up compared to a parameter estimation run using a ~ 500 CPU cluster. This implies a massive gain in cost-efficiency for GW parameter estimation.

4.4.2 Runtime Comparison

I measured wall-clock runtimes for various parameter estimation models defined in Table 1. The runtime of a nested sampling algorithm theoretically scales approximately linearly with the number of live points (N_{live}) and the number of MCMC steps per live point replacement (N_{steps}). I first verified this approximate scaling for my JAX implementation. Table 4 shows relative runtimes for Models A, B, and E with varying N_{live} and N_{steps} , normalised to a reference run.

Table 4: Verification of linear runtime scaling for Models A, B, and E (see Table 1). Runtimes are normalised relative to base cases of 10K live points, and $5 \times N_{\text{params}}$ MCMC steps.

Model	Live Points / 10K	Relative MCMC Steps	Relative Runtime
A	0.25	1	0.27
	0.75	1	0.72
	1.5	1	1.45
	1	2	2.14
	1	3	2.79
B	0.5	1	0.50
	1.5	1	1.51
E	0.5	1	0.53
	0.75	2	1.45

The data in Table 4 generally support the expected near-linear scaling with both N_{live} and N_{steps} for my JAX implementation. Based on this, I present a comparison of runtimes across different models and platforms in Table 5. A small, roughly constant compilation time is incurred by JAX runs at the start of each run. This means that for runs with small N_{live} , the measured runtime per live point might appear slightly higher. As N_{live} increases, this initial compilation cost becomes a smaller fraction of the total runtime, and the ‘Runtime / K Live Pts’ will approach its true asymptotic value.

The runtimes in Table 5 allow for key performance observations for the JAX-based pipeline:

- **Heterodyning:** Comparing identical JAX runs with and without heterodyning shows a dramatic speed-up. For GW150914, comparison of Models F and G the heterodyned F model is a factor of $22\times$ faster per K live points. For Models B and C, the acceleration from heterodyning is even more pronounced, yielding a speed-up factor of $900\times$. The larger impact on GW170817 is likely due both to its significantly longer signal duration (128s signal, 1024s PSD) compared to GW150914 (4s signal, 16s PSD), and the more complex waveform model (13 vs 11 parameters). These results clearly demonstrate the power of the heterodyning technique.
- **JAX vs Bilby:** A comparison between Model C and Bilby provides an informative comparison between the raw performance of GPUs vs. CPUs. Both of these models are unheterodyned and operate on the same volume of data. Rescaling the effective Bilby runtime to take into account the 4 extra parameters, I obtain an average wall-clock runtime of 36 hours for JAX and 1150 hours for Bilby. This implies a $32\times$ speed-up of a single L4 GPU compared to an equivalent single core CPU. This shows the raw hardware acceleration of GPUs. The key takeaway is that using my pipeline, a mid-tier GPU achieves runtimes that are dramatically faster than an equivalent single core CPU, and can be competitive with, and more efficient than, a large-scale CPU cluster. This represents a massive gain in throughput per unit of hardware, especially considering the cost analysis in Section 4.4.1.

Table 5: Comparison of parameter estimation runtime for various models I ran. The ‘Runtime rebased’ normalises to $N_{\text{steps}} = 5 \times N_{\text{params}}$. A dagger (†) implies rebased runtime. All times are in HH:MM:SS format. Models are defined in Table 1.

Event	Model	Live Points (K)	Runtime	Runtime † (rebased)	Runtime † / K Live Pts
GW170817	A	2.5	8:24	8:24	3:22
	A	7.5	22:38	22:38	3:01
	A	10	31:26	31:26	3:09
	A	15	45:29	45:29	3:02
	A (Flat in z)	10	59:51	29:55	3:00
	A (Uncertainty)	10	58:11	29:05	2:55
	B	5	12:02	12:02	2:24
	B	10	23:50	23:50	2:23
	B	15	35:55	35:55	2:24
	C (No Het)	1.216	44:19:57	44:19:57	36:27:27
	C (No Het)	1.3	45:58:59	45:58:59	35:22:17
	Bilby (456 CPUs)	1.216	4:37:32	4:37:32	3:48:14
	Bilby (532 CPUs)	1.596	3:49:17	3:49:17	2:23:39
	D	5	52:48	26:24	5:17
	D	10	55:00	55:00	5:30
	E	5	19:58	19:58	4:00
	E	7.5	54:52	27:26	3:39
	E	10	37:53	37:53	3:47
GW150914	F	5	10:06	5:03	1:01
	F	10	28:10	9:23	0:56
	F	10	18:54	9:27	0:57
	G (No Het)	5	3:28:30	1:44:15	20:51
	G (No Het)	10	3:35:59	3:35:59	21:35

- Event complexity and priors: As expected, runs with more parameters and more complex waveforms take longer per live point, due to the increased computational cost per likelihood evaluation. Runs with general priors also tend to be longer than those with constrained priors for the same event, as exploring a larger prior volume will require more iterations for convergence, as the likelihood landscape is wider and more complex.
- Impact of H_0 parameters: Comparing Model A (H_0, v_p) with Model B leads to a $1.27\times$ increase in runtime for A compared to B. Given the addition of 2 parameters, (a $15/13 \approx 1.15\times$ increase in dimensionality), and two extra Gaussian likelihood terms to evaluate, this runtime increase is reasonable. Similarly for Models D and E, we obtain a $1.5\times$ increase in runtime.

My JAX runtimes appear highly competitive and, for many configurations, are often significantly faster than other reported acceleration methods. For example, [Krishna et al. \(2023\)](#) estimate GW170817 parameters in 14 hours on a single CPU core with relative binning (1000 live points). Comparing this with my equivalent Model B, taking the 1,000 point runtime of 2:24, I achieve a $350\times$ improvement on their model, despite both runs using heterodyning.

4.5 Limitations

While my work demonstrates the significant potential of GPU acceleration for GW parameter estimation and its application to H_0 inference, I acknowledge several limitations:

- Waveform Model Availability and Accuracy: As discussed above, and in Section 3.1.1 and evidenced in the Results section, the current analysis relies on the waveform models available

within the `Ripple` library. Using more complete models that include orbital precession (e.g. `IMRPhenomXPHM`, `SEOBNR` family models) will yield better parameter estimates. The availability of a wide range of fully validated waveform models within the `Jim/Ripple` ecosystem is currently an area of active development. As these become more mature, incorporation into my pipeline would be a valuable extension to improve accuracy of parameter estimation.

- **Implementation Efficiency:** Although I achieved substantial speedups, the current implementation is not fully optimised. One specific area for improvement is the use of floating-point precision. I currently use double precision (64-bit) floating-point numbers for calculations, inherited from `Jim`. Experiments with the [Yallup et al. \(2025\)](#) sampler show that for CMB parameter estimation with 10,000 live points on the L4 GPU, we obtain runtimes of 00:13s for single and 1:40 for double precision. This demonstrates a speed-up of around $7\times$, which if implemented to my pipeline could further improve its acceleration factor.

5 Conclusion

In this project, I have presented the development, validation, and application of a GPU-accelerated pipeline for GW parameter estimation, constructed using the `JAX` framework. The primary scientific driver for this methodological development was to enhance our ability to perform rapid and robust Hubble constant inference using bright sirens, thereby contributing to ongoing efforts to resolve the Hubble tension. My work directly addressed the escalating computational demands of GW data analysis, particularly in anticipation of the large volumes of data expected from 3G detectors.

The central focus on H_0 inference guided my efforts. I successfully demonstrated that my GPU-accelerated pipeline can reproduce the landmark H_0 measurement from the bright siren GW170817, as originally presented by [Abbott et al. \(2017b\)](#). My derived value, $H_{0,\text{JAX}} = 70.4^{+14.9}_{-6.5} \text{ km s}^{-1} \text{ Mpc}^{-1}$, and is in excellent agreement with their findings. This underscores the scientific accuracy and cosmological utility of the pipeline I developed.

Furthermore, a key contribution of my work was a critical re-evaluation of the prior sensitivity analysis conducted by [Abbott et al. \(2017b\)](#) for their H_0 measurement. By leveraging the computational efficiency of my `JAX` pipeline, I was able to perform a direct nested sampling runs under alternative prior assumptions (specifically, a flat-in-redshift prior for the source distribution). My results indicate that the reweighting technique employed in the original study to assess the impact of this prior change likely underestimated its true effect due to limitations in posterior sample coverage. My direct sampling approach revealed a more significant sensitivity of the H_0 posterior to this prior choice than previously reported, highlighting the importance of robust methods for exploring prior dependence and the enabling power of fast inference tools like the one I developed.

Methodologically, I demonstrated the successful creation of a new parameter estimation pipeline built upon the `JAX` ecosystem, integrating components from libraries such as `Jim`, `Ripple`, and `BlackJAX`. A core design philosophy I adopted was to create a modular, flexible, and understandable framework. This was evidenced by the relative ease with which I could extend the pipeline to incorporate joint inference of H_0 and peculiar velocity v_p , and to implement different prior choices for sensitivity studies.

The performance gains achieved through GPU acceleration were also substantial. My pipeline demonstrated significantly faster runtimes on a single GPU compared to CPU-based methods, even when those methods utilised significant numbers of CPU cores, and other acceleration techniques. For instance, I showed that the L4 GPU could achieve analysis timescales competitive with, or more efficient in terms of total resource-hours than, large CPU cluster jobs. This makes complex analyses more tractable and timely, which is crucial for rapid multi-messenger follow-up of GW events – a key pathway to discovering more bright sirens for H_0 measurements. The improved cost-efficiency also makes advanced GW data analysis more accessible.

Looking forward, the pipeline I have developed can be further enhanced by incorporating more ad-

vanced waveform models as they become available in JAX-compatible forms, and by exploring further performance optimisations such as single precision computing. Its speed and flexibility make it a valuable tool for future GW data analysis, particularly for population studies of standard sirens which will be crucial for achieving high-precision H_0 measurements needed to definitively address the Hubble tension.

In conclusion, my work successfully demonstrates that GPU acceleration, harnessed through modern computational frameworks like JAX, offers a powerful and viable solution to the significant computational challenges in gravitational wave parameter estimation. The developed pipeline achieves substantial speedups while maintaining scientific accuracy, enabling complex analyses like joint H_0 inference on practical timescales. This capability is crucial for maximising the scientific return from current and future GW detectors and for advancing our understanding of cosmology.

6 Acknowledgements

During the writing process for this report, the AI language model Gemini was used only to assist with refining language and improving the clarity of expression. The core ideas, analysis, results, and conclusions remain entirely my own.

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